Assignment 5 - RNN vs GRU vs LSTM Comparison

Aim:

Compare SimpleRNN, GRU, and LSTM architectures for binary sentiment classification on the IMDB movie reviews dataset.

Concepts

- Recurrent Neural Network (SimpleRNN)
- Gated Recurrent Unit (GRU)
- Long Short-Term Memory (LSTM)
- Word embeddings
- Padding / truncating sequences to fixed length
- Binary sentiment classification
- Sigmoid output
- Binary cross-entropy loss
- Early stopping and learning rate reduction callbacks
- Dropout and L2 regularization
- ROC and Precision-Recall curves

Dataset

- IMDB movie reviews (Keras built-in dataset)
- Vocabulary: 10,000 most frequent words
- Sequence length: 200 tokens (padded / truncated)
- 25,000 train reviews, 25,000 test reviews
- Classes: negative (0) and positive (1)
- Balanced class distribution (50% / 50%)

Tools / Libraries

- TensorFlow / Keras (Embedding, SimpleRNN, GRU, LSTM, Dense, Dropout)
- Keras callbacks (EarlyStopping, ReduceLROnPlateau)
- scikit-learn (accuracy, precision, recall, F1, ROC-AUC, confusion matrix)
- NumPy, Pandas (result tables)
- Matplotlib, Seaborn (plots and heatmaps)

Models (High Level)

SimpleRNN

- Embedding (10,000 -> 64-dim vectors)
- SimpleRNN 64 units + Dropout
- Dense 1 + Sigmoid

GRU

- Embedding (10,000 -> 64-dim vectors)
- GRU 64 units + Dropout
- Dense 1 + Sigmoid

LSTM

- Embedding (10,000 -> 64-dim vectors)
- LSTM 64 units + Dropout
- Dense 1 + Sigmoid

Common Configuration

- Optimizer: Adam
- Loss: Binary cross-entropy
- Trained with early stopping and learning rate scheduling

Metrics

- Accuracy, Precision, Recall, F1-score
- ROC-AUC score
- Confusion matrix
- Training vs validation accuracy and loss curves
- ROC and Precision-Recall curves
- Comparative summary table across all three architectures

Methodology

```
IMDB Dataset
      |
Text Tokenization & Padding
      |
Build RNN / GRU / LSTM Models
      |
Train with Early Stopping
      |
Evaluate & Compare
      |
Visualize Results
```